

Ores

A naturally occurring substance – rock, sand, or crystal – that contains a large amount of an element is called an ore. Ores are mined so that the elements they contain can be removed. Some ores contain more than one element. The mineral malachite (right) is an ore of copper, and chemical reactions are used to extract the metal from the ore.



Malachite

Copper flakes reacting with nitric acid to form two compounds

The red-brown gas is nitrogen dioxide, a compound of nitrogen and oxygen.

Copper nitrate, a green, solid compound, in nitric acid



Compounds

Most elements do not stay pure in nature for long. Instead they react with other elements they come into contact with, which results in the atoms of two or more elements bonding together to make an entirely new substance called a compound. The properties of a compound are always very different to the elements that make it.

Mixtures

Elements combine chemically to form compounds, and it takes a chemical reaction to break the bonds between the atoms in a compound to separate the elements. On the other hand, a mixture is formed when compounds or elements are combined physically. An alloy is a mixture of at least two elements, one or all of which are metals. In a mixture, elements may be so thoroughly mixed that they are impossible to tell apart.



Mixture of food colouring and water